



EDUCATION ENHANCER USING AI

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PROJECT OVERVIEW

This project is an AI-powered education tool that generates interactive quizzes from user-provided content.

After taking the quiz, users receive automated feedback, performance summaries, and personalized study recommendations.





PROJECT SCOPE

- User Authentication & Authorization
- Quiz creation (Manual & with AI)
- Taking Quizzes by students
- Summarization & recommendation based on student level
- Preview students level (Quiz based) by the Instructor (admin)





USER AUTHENTICATION & AUTHORIZATION

- Instructor account is seeded once the first launch of the application
- Student can register to create accounts
- Forget-Reset password functionalities
- Email verification
- Email sending for reset and verification link



TECHNOLOGIES AND PLATFORMS USED

Application stack and technologies

- **React js** for the frontend
- **Node js** for the backend
- **Postgres** as database
- **Hugging Face inference**
- **Docker** for containerization
- **Github** for version control

During development:

- **Postman** for API testing and validation during development
- **Mailtrap** for email testing

Production technologies used for deployment:

- **Docker Compose:** Manage containers
- **Nginx:** Web server & load balancer
- **AWS (Cloud Services):** Online servers
- **GitHub Actions:** automated (CI/CD) pipelines



PROJECT FLOW

User (Frontend)
Enters a prompt

Backend

- Receives the prompt
- Sends it to the Hugging Face AI model

AI Model (DeepSeek R1)

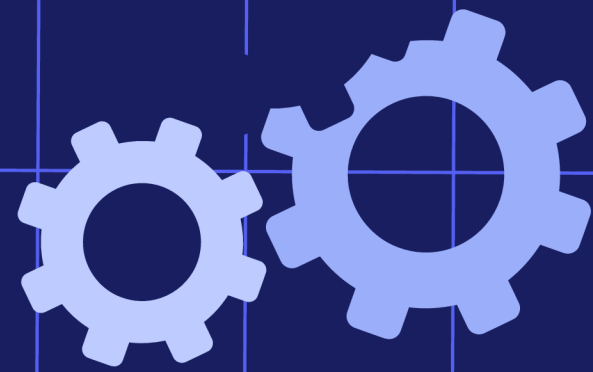
- Analyzes the prompt
- Generates a quiz:
 - Title
 - 10 Questions
 - 3 Answers per Question (1 correct)

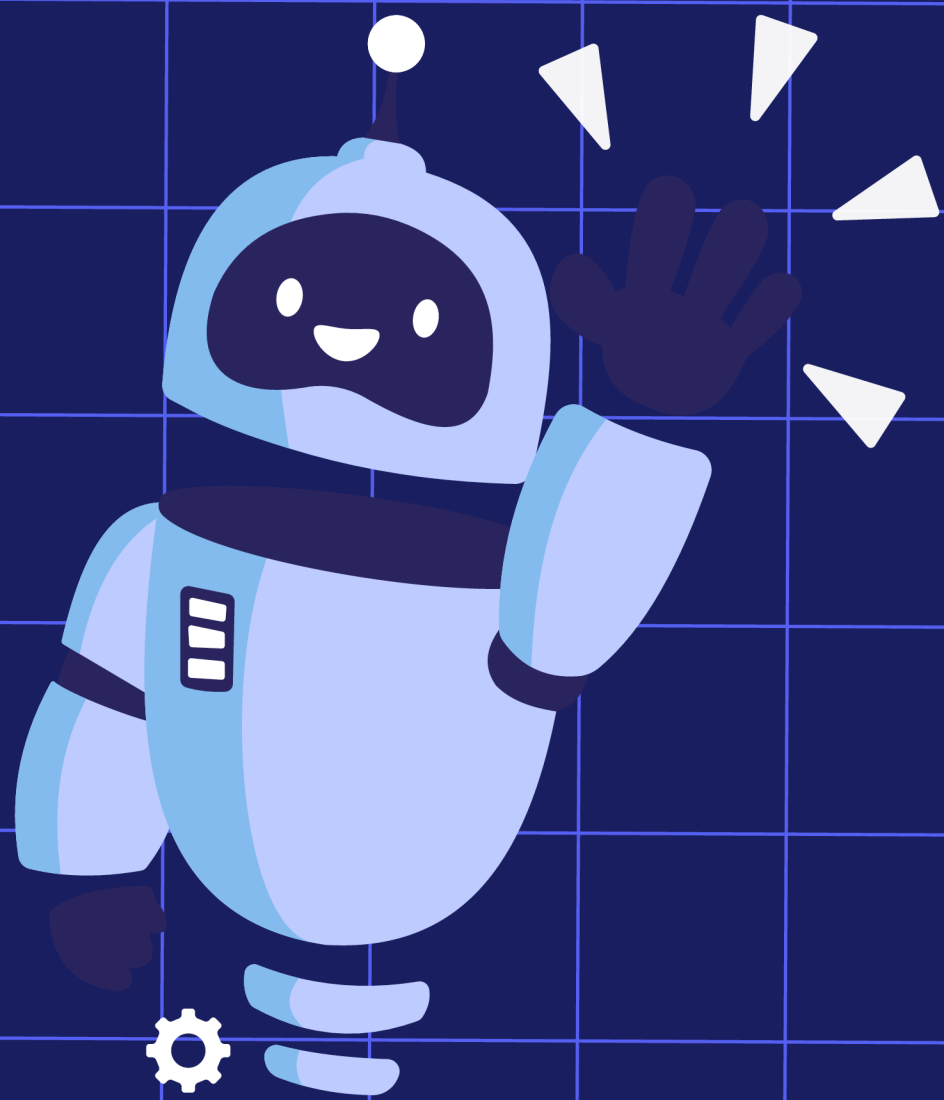
Backend again!

- Processes the quiz content
- Saves the quiz (title, questions, answers) to the database

Backend → Frontend

Sends the saved quiz to the user





DEMO TIME !

